

Time Series Week 1: Are you ready?

- What is the difference between cross-sectional and time series data?
- Can you define a time series?
- What is the general objective of a time series analysis, and how is that objective achieved?
- Why are standard regression models not appropriate to model time series data?
- What is the basic assumption underlying time series analysis/forecasting?
- What are the two main reasons we conduct a time series analysis in this course?
- What is autocorrelation at lag g in a time series?
- Why is autocorrelation a problem/challenge?
- Why is autocorrelation useful?
- When is a time series (process) stationary? Can you define a stationary time series?
- Can you identify whether a time series is stationary from a time series plot?
- What do time series plots enable us to do?
- What is a trend component? What gives rise to a trend in a time series? You should be able to recognise and describe (in detail) a trend component in a time series plot e.g. 'decreasing linear trend' not just 'trend'.
- What is a cyclical component? What gives rise to cyclical variation in a time series? You should be able to recognise cyclical variation in a time series, but also remember that we are not considering time series with cyclical variation in this course.
- What is a seasonal component? What gives rise to seasonal variation in a time series? You should know that seasonal variation due to the weather is just one example of seasonality. You should be able to describe the difference between seasonal variation and cyclical variation. You should understand what is meant by the length of a seasonal variation and the number of 'seasons' in one seasonal variation.
- What is random/irregular variation? What gives rise to random variation in a time series? When is random variation problematic? Do all time series have random variation?
- Given a time series plot, can you identify the components present that are influencing the pattern we see in the data?
- What is an additive time series model? Can you express it mathematically?
- What is a multiplicative time series model? Can you express it mathematically? Why is the multiplicative model preferred to the additive model (in this course)?
- When is moving average smoothing used? There are two main uses for it – can you describe these two uses? You will not be expected to apply $MA(k)$ to an entire time series but are expected to be able to calculate a few $MA(k)/CMA(k)$ values.
- What is the advantage of moving average smoothing? What are the disadvantages of moving average smoothing?
- What is the appropriate value of k to use to smooth out seasonality from a time series that has seasonality? Why would you want to smooth out the seasonality?
- Why do we apply classical decomposition to a time series? You will not be expected to apply a classical decomposition to an entire time series but should be able to compute a few missing values in output from a classical decomposition model AND be able to explain the overall process. For example, how we get our seasonal indices and why they often don't sum exactly to the number of seasons in a seasonal variation.
- How does one correctly interpret an adjusted seasonal index? You should be able to understand and interpret adjusted seasonal indices in the context of the example you are given.

Time Series Week 2: Are you ready?

- Why do we consider simple forecasting methods?
- Can you describe each of the four simple forecasting methods?
- Can you identify which simple forecasting method will give the most accurate short-term forecasts for a given time series? Can you explain why?
- Can you identify which simple forecasting method will give the most accurate long-term forecasts for a given time series? Can you explain why?
- What is 'h', the 'forecast horizon'?
- Can you identify a stationary or a non-stationary time series from a time series plot?
- Can you explain why a time series with a trend and/or seasonality is non-stationary?
- Can you identify non-constant variance in a time series plot?
- What are the family of variance-stabilising transformations called?
- If you were given RStudio output with the value of `lambda` from a `BoxCox.lambda()` function of 0, would you be able to describe what variance-stabilising transformation was applied? What about a value of 0.25 or 0.33 or 0.5?
- Can you decide whether a variance-stabilising transformation has worked by comparing time series plots of before and after the transformation?
- Can you describe what transformation would be appropriate to remove a linear trend? Both name the transformation and describe mathematically how it is achieved?
- Can you describe what transformation would be appropriate to remove seasonality at lag 'g'? Both name the transformation and describe mathematically how it is achieved?
- Given a non-stationary time series plot, would you be able to describe in detail why it is not stationary and then identify and describe what transformations are required to make it stationary?
- What does an ACF (correlogram) plot show you?
- What do the dashed blue lines on an ACF plot represent?
- If there are non-zero autocorrelations in the ACF of a time series, what does this mean?
- If there are non-zero autocorrelations in the ACF of the residuals of a fitted model, what does this mean?
- What does one want the ACF of the residuals of a fitted model to look like? Why?
- If you were shown RStudio output of the functions `ndiffs()` or `nsdiffs()` would you be able to interpret the output? e.g. you get a value of 1 from the `ndiffs` function applied to a time series – what does this mean about the time series, and what does it mean about the next step for you to take in your time series analysis?
- Can you clearly state the four assumptions/properties that the residuals of a fitted model should display? Can you explain what autocorrelation in the residuals of a fitted model means about that model?
- Can you conduct a Ljung-Box test for autocorrelation on either time series data or on the residuals of a fitted model? Do you know what the difference in the degrees of freedom of the test statistic would be between these two uses of the test? Do you know how to use the Chi-square distribution to evaluate the strength of the evidence (represented by the test statistic) against the null hypothesis?
- Why do we calculate measures of forecast accuracy?
- Can you explain the difference between MAE and MSE measures of forecast accuracy? When would you use MSE instead of MAE?
- What value for these measures would you like to see for a fitted model? Can you calculate these values given an example with a few observations?
- Why do we split data into a training set and a test set? On which data do we fit our model?

- What proportion of data (give a range) do we usually use as a training set? What proportion of data (give a range) do we usually use as a test set?
- Why is it important that the model never sees the data in the test set?
- Why do we ignore the training set accuracy measures that we get from running the `accuracy()` function in RStudio?
- Given some output, can you use the appropriate measures of forecast accuracy to compare a few fitted models and choose the best model to forecast future values for a time series?
- What is a prediction interval? Why do we need to compute them? Do you expect the width of a prediction interval to increase or decrease the further into the future you predict?
- Given the necessary values, can you calculate a prediction interval?

Time Series Week 3: Are you ready?

- Can you explain how exponential smoothing models model a time series?
- Can you explain how exponential smoothing models model the components of a time series?
- Can you explain/interpret the effect of the magnitude of the smoothing constants?
- Can you identify which exponential smoothing model is appropriate to fit to a time series?
- Do you understand the information in the RStudio output of a fitted exponential smoothing model?
- Given the relevant formulae and information, can you calculate some values of the different components? Can you forecast some values (both within the observed time series and into the future)?

Time Series Weeks 4 & 5: Are you ready?

- Can you define/explain what is autocorrelation at lag 'g'?
- Can you define/explain what is partial autocorrelation at lag 'g'?
- Can you interpret the value of the autocorrelation/partial autocorrelation at a particular lag?
- Can you explain why autocorrelation is both a problem and is important in the context of time series analysis?
- Can you explain what positive/negative autocorrelation looks like in a time series?
- Can you clearly define what is a stationary time series?
- Given a plot of a time series, can you explain why the time series is stationary/non-stationary?
- Can you describe two uses of the correlogram (sample ACF) plot?
- Can you identify what components would be present in a time series based on just the behaviour of a correlogram?
- Can you define a white noise process? Can you explain why we would want the residuals of a fitted model to follow a white noise process?
- Can you explain what the dashed blue lines on a correlogram/partial correlogram represent? Do you know how to interpret ACF/PACF values that fall outside of the dashed blue lines vs those that fall inside the dashed blue lines?
- Do you understand how to interpret what a 95% confidence interval means in terms of the number of ACF/PACF lines you would expect to see outside the dashed blue lines?
- Can you explain why a white noise process is always stationary, but a stationary process is not necessarily white noise?
- Can you explain the behaviour of the correlogram for a time series with trend, with seasonality or with both trend and seasonality?
- Given a correlogram, can you identify what components would be present in the time series (without having a time series plot)?

- What is an ARIMA(p,d,q) model? Can you explain to what the AR, I and MA parts refer? Can you explain to what the p, d and q refer?
- Can you explain what information ARIMA models try to model in a time series?
- Can you explain the difference between exponential smoothing models and ARIMA models in how/what they model in a time series?
- Can you list the steps that are followed in the general approach to modelling a time series when we are focusing on autocorrelation i.e. with an ARIMA model?
- Can you explain what distribution the error series follows in an AR/MA/ARIMA model, and why it is assumed that it follows this distribution?
- Can you explain how an AR(p) model models a time series?
- Can you state the (mathematical) formulation of an AR(p) fitted model?
- Can you state the mathematical formulation on an AR(1) model?
- Can you explain the behaviour of an AR(1) model for values of the coefficient close to 1 or close to -1?
- Given the appropriate formulae and values in a fitted AR(1) model output, can you calculate the mean, constant term, variance and autocorrelation of an AR(1) model?
- Given the appropriate formulae and values in a fitted AR(p) model's output, can you calculate the mean of an AR(p) model?
- Can you explain the combined behaviour of the ACF and PACF for an AR(p) model?
- Can you explain how a MA(q) model models a time series?
- Do you understand the difference between a MA(q) model and MA(k) smoothing?
- Can you explain the difference between how AR(p) and MA(q) models model a time series?
- Can you state the (mathematical) formulation of a MA(q) fitted model?
- Given the appropriate formulae and values in a fitted MA(q) model's output, can you calculate the mean and variance of a MA(q) model?
- Can you explain why MA(q) models are always stationary?
- Can you explain the combined behaviour of the ACF and PACF for a MA(q) model?
- Can you state the mathematical formulation on a MA(1) model?
- Given the appropriate formulae and values in a fitted MA(1) model output, can you calculate the mean variance and autocorrelation of a MA(1) model?
- Can you explain the combined behaviour of the ACF and PACF for when an AR(p) i.e. ARIMA(p,d,0) model is appropriate to fit to a time series?
- Can you explain the combined behaviour of the ACF and PACF for when a MA(q) i.e. ARIMA(0,d,q) model is appropriate to fit to a time series?
- Can you explain the two cases of combined behaviour of the ACF and PACF for when a combined AR-MA i.e. ARIMA(p,d,q) model is appropriate to fit to a time series?
- What are information criteria? What do they measure? How do we use them in a time series analysis?